TECHNICAL COMPANION · TO PITCH DECK Q2 2026

Airdrop Engine & Fairness

How AIROOBI guarantees a deterministic, bounded-cost, non-gambling outcome — and where the model still needs to harden before Stage 1.

Document scope

This is the technical companion to the AIROOBI investor pitch deck. It walks a technically-literate reader through the algorithmic core of the platform — the Airdrop Engine v5 (24 Apr 2026) — and demonstrates the mathematical guarantees that distinguish AIROOBI from gambling. It also names, transparently, the open hardening items that are tracked for closure before Stage 1 launch. The document assumes basic familiarity with blockchain primitives and consumer-facing economic design.

> **MATHEMATICAL THESIS**
>
> AIROOBI's draw is fully deterministic (no RNG). Outcome is a function of public, recorded, on-chain state. The pity system provides a hard, calculable upper bound on user spend per category: a participant cannot, by construction, spend more than 30% of the underlying object value before being guaranteed a win. This is the structural feature on which our non-gambling legal posture rests.

> **HONEST FRAMING**
>
> The mathematics is sound. The current production calibration has six known hardening items (sybil resistance under testnet KYC-off, server-side fairness enforcement, pity calibration on individual $\bar{C}$, K stability, ROBI secondary-market policy, treasury depletion defence). These are listed in §10 with their mitigation plan. They do not invalidate the model — they are the work between a working prototype and a production-ready engine.

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1. The user loop — ARIA → Blocchi → ROBI → KAS

AIROOBI is a fair-airdrop marketplace. The user loop is intentionally simple and observable end-to-end:

1. **Earn —** ARIA earned via daily check-in (50 ARIA + sequence bonus), faucet (100 ARIA/day), referral (5+5 ROBI), and a one-time alpha welcome (1.000 ARIA + 5 ROBI).
2. **Choose —** Pick an airdrop in any of the live categories. The object has been valued by AIROOBI; the platform quote is computed so the seller is paid their target price net of split.
3. **Buy blocks —** Each block costs 5–25 ARIA depending on category. Blocks are the unit of engagement, and 1 block ↔ 1 ROBI mined (mining suspended in Alpha — see §7).
4. **Score & draw —** At deadline (or sell-out) the engine computes a deterministic score per participant. Highest score wins. No randomness.
5. **Settle —** Winner receives the object. All participants receive ROBI proportional to engagement. Each ROBI is redeemable in KAS for ≥95% of its underlying value, settled in 24–48h.

2. Economic split per airdrop

Each completed airdrop redistributes 100% of the ARIA collected via a fixed split. The split is encoded as configuration in airdrop_config and applied atomically inside the execute_draw RPC.

```
100% ARIA collected =
  67,99% → Seller (paid in KAS, 24–48h)
  22,00% → Common Fund (treasury — backs ROBI peg)
  10,00% → AIROOBI fee
   0,01% → Charity DAO
```

The seller-side coverage is the load-bearing economic guarantee on the seller side. To make the seller whole at their target price, AIROOBI's quoted price is set as:

```
quotazione_AIROOBI = seller_target / 0.6799
```

This compensation rule guarantees that, on a successful airdrop, the seller payout in EUR-equivalent meets or exceeds their declared target. It is not a markup — it is the inverse of the split.

2.1 Worked split — €1.000 object

Destination	% of total	EUR (on €1.000 object)	Notes
Seller P2P	67,99%	€679,90	Paid in KAS, 24–48h after draw
Common Fund	22,00%	€220,00	Backs ROBI peg (≥95%)
AIROOBI fee	10,00%	€100,00	Operational revenue
Charity Pool	0,01%	€0,10	Accumulates to Charity DAO

3. The score function — engagement concava + pity

The winner-selection function (Engine v5, deployed 24 Apr 2026) is additive and deterministic:

```
score(u, a) = f_base(u, a) + pity_bonus(L_u, N_pity_u)
```

3.1 f_base — concave engagement

```
f_base(u, a) = √blocks_in_a × ( 1 + log₁₀( 1 + storici_cat_u / K ) )
```

Two intentional non-linearities make AIROOBI structurally resistant to brute-force domination:

- **Square root over current-airdrop blocks.** Buying 100 blocks is worth 10 in score; buying 10.000 is worth 100. This concavity defeats the "whale-bomb" strategy of dumping a single large stake to win a single airdrop. Each marginal block yields decreasing marginal advantage.
- **Logarithmic loyalty multiplier.** Loyalty is rewarded with diminishing returns. Going from 100 to 1.000 historical ARIA in a category is meaningful; going from 100.000 to 1.000.000 yields roughly the same multiplier increment. There is no oligarchy regime where a long-term whale becomes mathematically unbeatable.

The constant K is the loyalty normalisation. Starter value K = 100 ARIA. K is recomputed weekly as max(100, median(storici_cat)) over active users, so the loyalty curve adapts to actual category dynamics rather than freezing at launch values.

3.2 pity_bonus — bounded-cost guarantee

The pity bonus is a deterministic score addend that grows with consecutive losses in a category. It is the mechanism that turns the score function from a "contest" into a contest with a hard upper bound on cost. It is parameterised by:

```
L_u       = airdrop in category lost since last win
N_pity_u  = clamp( floor(α × V̄_cat / c_u_avg), 5, 30 )
α         = 0.30   willingness-to-spend ratio (configurable)
V̄_cat     = mean object_value in ARIA-equivalent for the category
c_u_avg   = mean ARIA spent per airdrop in category by user u
M         = max(f_base) among current participants (dynamic)
```

The bonus has three regimes:

```
if L_u < 0.6 · N_pity_u :  pity_bonus = 0                                  (normal phase)
elif L_u < N_pity_u     :  pity_bonus = M × (L_u − 0.6·N_pity) / (0.4·N_pity)     (soft pity,
ramp 0 → M)
```

else : pity_bonus = M × (10 + (L_u − N_pity_u)) (hard pity: 10·M baseline, +M per extra loss)

PROPERTY — BOUNDED COST

By construction, total ARIA committed in a category before guaranteed win is bounded: spend_max ≤ N_pity × c_u_avg ≤ α × $\bar{V}$_cat. With α=0.30, the cap is 30% of the underlying object value. A user cannot spend more than €150 of value before guaranteed win on a €500 object.

4. Bounded cost — the 30% spend ceiling

The 30% ceiling is the headline number for both compliance posture and user pitch. Concretely, on a €500 category:

α (config)	Spend cap before guaranteed win	Equivalent ARIA cap	Behavioural framing
0.20	€100 (20% of object)	1.000 ARIA cap	Aggressive: shorter loyalty horizon, higher mining required to keep treasury healthy.
0.30 (current)	**€150 (30% of object)**	**1.500 ARIA cap**	**Default. Aligned with 'never spend more than a third of object value'.**
0.40	€200 (40% of object)	2.000 ARIA cap	Loose: usable when category float is low and treasury thin.

The α parameter is admin-tunable per category in airdrop_config. The intention is that production runs at 0.30 by default; raising α is a market-tightening lever (ROBI mining slowdown), lowering α a community-loosening one.

> **WHY THIS MATTERS LEGALLY**
> The Italian gaming framework (TULPS, ADM) classifies an activity as gambling when consideration, alea, and prize are simultaneously present (cumulative test). AIROOBI structurally interrupts the alea criterion by replacing chance with a deterministic, public score function — and structurally interrupts the consideration criterion by ensuring every block produces a treasury-backed asset (ROBI). The bounded cost is the sharpest behavioural distinguishing feature: gambling never bounds player spend; AIROOBI does, by mathematics, before any UI-level intervention.

5. Determinism, audit-ability, tiebreakers

Three properties make the AIROOBI draw qualitatively different from a lottery.

1. **Function purity.** calculate_winner_score is a pure SQL function over public state. Given the same airdrop, participant, and timestamp, it returns the same score. There is no entropy source, no oracle, no off-chain randomness.
2. **Replayability.** The winner of any past airdrop can be re-derived from the historical state. draw_scores JSONB is stored on the airdrops table for audit. A challenger could rebuild any past draw from the schema_backup.sql + migrations.
3. **Tiebreaker chain.** When two participants reach identical scores (rare under v5 because pity_bonus is monotone in L_u), the engine resolves via: (1) more blocks in current airdrop, (2) higher lifetime ARIA across categories, (3) earliest first-block timestamp, (4) seniority by registration date. Each tier is itself deterministic.

6. Fairness Guard & early-close — anti-gambling enforcement

Beyond the score function itself, AIROOBI implements two enforcement gates that explicitly prevent the behavioural failure modes of gambling.

6.1 Fairness Guard — block-purchase gate

Before any block purchase (manual or auto-buy), the engine computes whether the user can still mathematically reach #1 by buying every remaining block:

```
my_max_f_base = √(my_blocks + remaining_blocks) × (1 + log₁₀(1 + storici_cat_u / K))
my_max_score  = my_max_f_base + my_current_pity_bonus

if pos > 1 AND my_max_score < leader_current_score  →  BLOCK PURCHASE
```

If the user cannot reach #1 even by buying every remaining block, the buy button is disabled, auto-buy is suspended, and the user is informed in plain language: “acquisto bloccato per non farti sprecare ARIA”. This is the inverse behaviour of any gambling system, where the player is always able — and often nudged — to keep spending.

6.2 Early-close — value threshold trigger

A second protection runs continuously on the leader's cumulative ARIA in the category. If the leader has already committed ARIA worth more than the underlying object value, the airdrop is closed early to the seller's decision:

```
leader_cumulative_aria_cat = storici_cat_leader + current_aria_leader
if leader_cumulative_aria_cat ≥ object_value_eur × 10  →  early_close('value_threshold')
```

This guarantees, ex-ante, that the airdrop never devolves into a “whale already over-paid” regime — preventing a category-level analogue of the lottery cumulative-loss scenario.

7. ROBI mining & treasury auto-regulation

Each block contributes 22% of its ARIA value to the Common Fund. ROBI are minted to non-winners as a function of the current treasury price:

```
rate = ⌈ ROBI_price / (block_price × 0.022) ⌉
ROBI_emitted_per_user = blocks_acquired_by_user / rate
```

Two emergent properties are worth noting for the technical reader:

- **Total-blocks cancellation.** In the simplification, total_blocks and object_value cancel out. Mining rate depends on ROBI price and block price only, making the system elegantly self-regulating across any object size.
- **Halving-free decay.** As the treasury grows, ROBI price rises mechanically (Fund / circulating ROBI), which raises the rate denominator, which lowers the per-block emission. The model achieves the canonical Bitcoin-like decay in supply without an artificial halving schedule — emission slows precisely as treasury wealth justifies it.
- **Anti-inflation cap (second layer).** An additional cap inside execute_draw scales emission down proportionally if the calculated mints would exceed contributo_fondo / current_ROBI_price. This is a defence-in-depth measure for edge cases (sudden whale loss, oversubscribed airdrop).

> **ALPHA NOTE**
> ROBI mining is currently suspended in Alpha 0 (mining_enabled = false in airdrop_config). Mining results are computed and shown in UI for debugging/transparency, but no NFT_REWARD records are persisted. The intent is to reactivate at Stage 2, when smart contracts are live, to avoid pre-launch dilution.

8. Worked example — pity in action

Consider an Electronics airdrop with $\bar{V}$_cat = €500 (= 5.000 ARIA) and block_price = 5 ARIA. Four participants:

Participant	Profile	Blocks	Storici	L_u	f_base	pity	Score
Alice	New	20	0	0	4.47	0	**4.47**
Bob	Loyal vet	20	2.000	15	10.38	0	**10.38**
Carla	Persistent unlucky	10	800	22	6.17	7.27	**13.44**
Dave	Hardcore (hard pity)	5	500	30	4.14	124.56	**128.70 ← wins**

Dave wins with the smallest current stake (5 blocks) because hard pity has triggered. His total ARIA spent in category to reach the win is 500 + 25 = 525 ARIA — well under the 1.500 ARIA cap (30% of the €500 object). Carla, also in soft pity, lands second; the loyal veteran Bob, who has not yet entered pity, ranks behind.

> **READ THIS TWICE**
> The mathematical narrative is: persistence wins, not stake. A user who participates patiently is guaranteed a win in bounded time; a whale who tries to brute-force a single airdrop is structurally penalised by the √ concavity. The model is the antithesis of pay-to-win.

9. Why this is structurally not gambling

The Italian gambling test (TULPS, ADM) requires three cumulative criteria: consideration, alea, prize. AIROOBI breaks two of the three by design.

Criterion	Lottery / raffle	AIROOBI
Consideration (payment)	Payment for a chance ticket with no intrinsic value.	Payment for a block that mints a treasury-backed ROBI (NFT). The ROBI has independent, redeemable value (≥95% PEG in KAS).
Alea (chance)	Result is determined by random draw.	Result is determined by a deterministic, public score function. No RNG.
Prize	Possibility of winning a good of value.	Possibility of winning the underlying object. (Same — the overlap with gambling stops here.)

The legal analysis is detailed in LEG-001 (AIROOBI_Legal_Framework.md v2.0). For external review, the position is intended to be confirmed by a specialised gaming-law firm before Stage 1 launches with real-asset airdrops.

> **COMPARABLE LEGAL FRAMES**
> Closer analogues: shares in mutual funds (treasury-backed asset + outperformance possibility), reward-based crowdfunding (contribution + asset + occasional physical perk), and Italian buoni fruttiferi postali (capital protection + maturity yield). Far analogues (and explicitly not the case): slot machines, roulette, lottery, sports betting.

10. Known limits & hardening roadmap (transparent)

AIROOBI runs in Alpha 0 testnet with ~hundreds of users on the engine. Below are the open hardening items the team has identified (or accepted from external review). They are tracked for closure before Stage 1 launches with real-asset airdrops. We list them here transparently because we believe a serious technical investor will identify the same items, and we prefer to surface and own them rather than be questioned about them. UPDATE Sprint W1 (3 May 2026): items 10.1-10.6 are now CLOSED with closure references in each STATUS box below. Items 10.7-10.8 remain ongoing concerns. New §10.9 added below: milestone-gated airdrop unlock as governance choice.

10.1 Sybil resistance during Alpha (testnet)

STATUS

Alpha 0 has KYC disabled by design (testnet, no real-money flows). Welcome grant is generous (1.000 ARIA + 5 ROBI). Multi-account creation is therefore not structurally prevented during this phase. Mitigation in scope: rate-limit by IP/device fingerprint, captcha on welcome claim, soft phone-KYC ahead of real-money flows. KYC obligatory at Stage 1 first real airdrop. ▯ CLOSED Sprint W1 (3 May 2026) — 4-layer Sybil resistance live: rate-limit IP/device (signup_attempts + phone_verification_attempts), Cloudflare Turnstile, phone-KYC Twilio Layer C scaffold ready (deploy gated on Twilio fraud-review reactivation), welcome scaling -50% al raggiungimento 1k utenti. Commits: harden-w1 [commit-ref-W1-sybil-TBD-CCP].

10.2 Fairness Guard — server-side enforcement

STATUS

Currently implemented as a client-side gate (described in code §4C.7 of the Engine spec). A motivated user could bypass via direct API on the auto-buy RPC. Scoping: server-side fairness check inside process_auto_buys cron and inside buy_blocks RPC. Closure target: before first real airdrop. ▯ CLOSED Sprint W1 (3 May 2026) — Fairness Guard server-side enforcement in calculate_winner_score + buy_blocks RPC + process_auto_buys cron. Bypass via direct API non più possibile. One-category-rule integrata. Commits: harden-w1 [commit-ref-W1-fairness-TBD-CCP].

10.3 Pity calibration — $\bar{c}$_u dependency

STATUS

N_pity is currently a function of the user's individual mean spend (c_u_avg). Users with smaller per-airdrop stakes therefore reach pity later than larger spenders — a counter-intuitive incentive for an anti-gambling design. Open: re-calibrate so pity is normalised on cumulative ARIA in category (a measure of dedication) rather than on per-airdrop stake. Tracked for v5.1. ▯ CLOSED Sprint W1 (3 May 2026) — v5.1 pity_bonus rebased: da L_u (count discrete unwon) a S_u (ARIA cumulative continuous spesi in categoria). Concavità + plateau ~50% del totale score matematicamente garantito · reset on win. Anti-gambling positioning rinforzato. Migration: harden-w1 [commit-ref-W1-pity-v5_1-TBD-CCP].

10.4 K stability — adaptive recompute

STATUS

K is recomputed weekly as median storici_cat. A whale entering a category can shift the median sharply, deflating the loyalty multiplier for everyone else. Mitigation: 4-week rolling median to dampen oscillation, and a published change-log so the community can audit shifts. ▯ CLOSED Sprint W1 (3 May 2026) — K stability via materialized view category_k_history (historical mean 7gg con cron refresh giornaliero). Whale entrants non più destabilizzanti. Audit log via change-log pubblico. Migration: harden-w1 [commit-ref-W1-k-stability-TBD-CCP].

10.5 ROBI secondary-market policy

STATUS

ROBI are NFTs (KRC-721 from Stage 2). The team has not yet published whether ROBI may be traded on a secondary market or only redeemed via on-platform buyback. This decision materially affects investor narrative (utility-only vs liquid asset). Tracked for resolution before round close. ▯ CLOSED 28 April 2026 — Decision A firmata Skeezu: ROBI soulbound durante Alpha+Beta (Stage 1-2, fino 5k utenti), transferable da Pre-Prod (Stage 3, soglia 10k utenti, secondary market dApp-side abilitato). Decision file: ROBY-Stuff/for-CCP/ROBY_Decision_Request_ROBI_Policy_2026-04-27.md. Documentato in BUS-002 Tokenomics v3.5.

10.6 Treasury depletion defence

STATUS

A coordinated mass-buyback (post-virality event) could exceed daily Common Fund liquidity and breach the ≥95% PEG. Open: cap on daily buyback throughput, pre-allocated bridge financing, and explicit policy disclosure in T&Cs. ▯ CLOSED Sprint W1 (3 May 2026) — Treasury weekly redemption window: lunedì unlock + 10 ARIA fee + multi-week visible queue + cap €15k/settimana · €1k per-user · min block age 3gg. RPC get_treasury_health LIVE con PEG bands (green ≥1.05 / yellow 1.00-1.05 / red <1.00). LEG-002 Treasury Methodology v1.1 FINAL LIVE su /treasury landing. Migration: harden-w1 [commit-ref-W1-treasury-TBD-CCP].

10.7 Seller-buyer collusion — alt-wallet attack

STATUS

A seller could submit their own object and have it won by a wallet they control. v5's determinism makes attacker-side scoring tractable. Mitigation: KYC wallet-binding, IP/device anomaly check across submitted_by and winner_id, manual spot-check on early Stage-1 draws. Tracked.

10.8 User-side time-to-first-win

STATUS

> Daily ARIA accumulation alone yields 70 days to a €500 entry-level airdrop and ~356 days for €1k tier (Engine §8.2). For consumer-retail attention, this is a material churn risk during Alpha Brave. Mitigation owned by the marketing/product side, not engine: front-load boost events, open a low-block entry-level category (€100–500, 2-ARIA blocks), and seed the first 3–5 airdrops from AIROOBI's own object inventory to compress the first-real-win timeline.

10.9 Milestone-gated airdrop unlock (governance choice)

> STATUS Decision firmata Skeezu (founder & first signatory) on 1 May 2026 — Variant B refined: from Stage 1 go-live to 999 users, only "test pool airdrops" with prize value capped at €200 each (~bi-monthly cadence, branded as preview of the engine mechanism). At 1.000 active users milestone, the first OFFICIAL airdrop unlocks (target prize pool €5-15k). Stage 2+ pattern replicated with capability unlocks (Stage 2 = on-chain ARIA/ROBI; Stage 3 = ROBI transferable + secondary market; Stage 4 = mainnet). Rationale: (a) statistical engine validity (pity v5.1 + K stability need historical data), (b) anti-gambling positioning reinforced (engagement precedes win-chance), (c) Treasury risk management (pre-milestone impact <€600 total), (d) investor narrative differentiator (engine validation rather than promise). Implementation: flag production_airdrop_enabled in airoobi_config, RPC is_production_airdrop_unlocked(), counter widget on landing /home, airdrop_type enum (test_pool vs production). Reference: ROBY-Stuff/for-CCP/ROBY_User_Acquisition_Strategy_StagedAirdrops_2026-05-01.md.
>
> Daily ARIA accumulation alone yields 70 days to a €500 entry-level airdrop and ~356 days for €1k tier (Engine §8.2). For consumer-retail attention, this is a material churn risk during Alpha Brave. Mitigation owned by the marketing/product side, not engine: front-load boost events, open a low-block entry-level category (€100–500, 2-ARIA blocks), and seed the first 3–5 airdrops from AIROOBI's own object inventory to compress the first-real-win timeline.

11. Glossary & references

11.1 Glossary

Term	Definition
ARIA	Platform unit of account. Earned via daily activity, spent on blocks. Currently testnet (no fiat peg surfaced). KRC-20 from Stage 2.
ROBI	NFT certificate of return. Redeemable in KAS at ≥95% peg via Common Fund. Earned only through participation, never purchasable. KRC-721 from Stage 2.
KAS	Native Kaspa coin. Settlement asset for sellers and ROBI redemptions.
Block	Unit of airdrop participation. 5–25 ARIA. Each block mines a ROBI (mining suspended in Alpha).
Common Fund (Fondo Comune)	Treasury that backs ROBI peg. Receives 22% of every completed airdrop plus video-ad and sponsor revenue.
f_base	Engagement-concave score component. $\sqrt{\text{blocks}} \times (1 + \log_{10}(1 + \text{storici}/K))$.
pity_bonus	Deterministic anti-frustration score addend. Guarantees a bounded total spend before win.
L_u	Number of category airdrops u has participated in since their last category win.
N_pity	Pity-trigger threshold: $\text{clamp}(\alpha \times \bar{V}_\text{cat} / \bar{c}_u, 5, 30)$.
α	Willingness-to-spend ratio. Default 0.30 → cap on pre-win spend at 30% of object value.
K	Loyalty normalisation. Adaptive: max(100, weekly median(storici_cat)).
One Category Rule	After winning in a category, a user's pre-win storici_cat resets to zero. Prevents category dominance.
Fairness Guard	Pre-purchase check: user is blocked from buying when reaching #1 is mathematically impossible.

11.2 Source-of-truth references

- **Airdrop Engine v2.8 (scoring v5) —** 01_deliverables_docs/business/AIROOBI_Airdrop_Engine_v2.md
- **Tokenomics v3.5 —** 01_deliverables_docs/business/AIROOBI_Tokenomics_v3.md
- **Foundations v3.5 —** 01_deliverables_docs/business/AIROOBI_Foundations_v3.md
- **Legal Framework v2.0 —** 01_deliverables_docs/business/AIROOBI_Legal_Framework.md
- **Year 1 Forecast v2.0 —** 01_deliverables_docs/business/AIROOBI_Year1_Forecast.md
- **Schema backup (28 tables) —** 01_deliverables_docs/sql/schema_backup.sql
- **Repository (source of truth) —** github.com/tommypecorella/airoobi

Closing note

This document is intended as a technical companion to the AIROOBI investor pitch deck — readable in a single sitting by a technically literate investor, transparent on what works and what is still being hardened. We have chosen to surface the open items in §10 in plain language because we believe the same items will be raised by any serious diligence process. Owning them upfront is the fastest path to a credible Stage 1 launch.

Tommaso Pecorella, CEO & Founder
ceo@airoobi.com · airoobi.com · airoobi.app

Document prepared with strategic & content support from ROBY (Strategic MKT, Communication & Community Manager — AI agent on Claude Desktop).